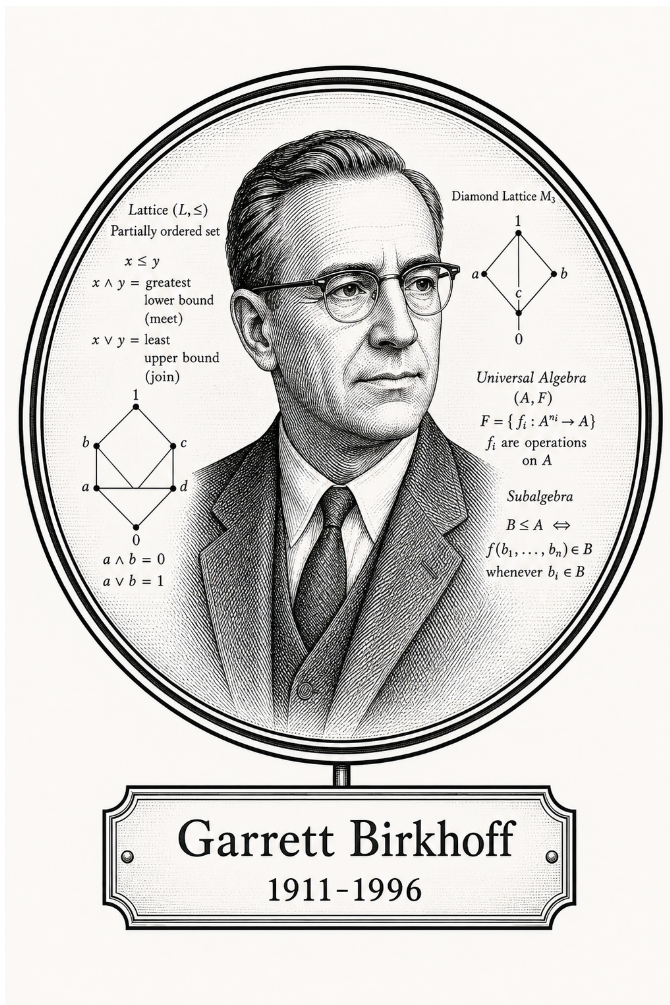
From Cantor to Ito

Algebra

Lattice and Order Theory



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Self-Study Notes

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For every reader learning to make mathematics their own.

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1.1 Breadcrumb

Why Lattice Theory Gets Its Own Home

Remark (Why lattice theory gets its own home). Lattice theory begins with a simple observation: in many ordered structures, two elements have a natural least upper bound and a natural greatest lower bound. Once those operations exist everywhere, order starts behaving like algebra.

Remark (Where the first pass already lives). The first introduction to lattices is already in the lattice section of the order notes. That is the right place for the first encounter, because it grows directly out of partial orders, suprema, and infima.

Remark (Why return later). Once lattices stop being examples and start becoming the main character, the story gets richer. A fuller lattice-theory chapter will naturally gather topics such as:

- distributive, modular, and Boolean lattices;
- complete lattices and closure operators;
- sublattices, ideals, and filters;
- Galois connections and adjoint situations;
- fixed-point theorems such as Knaster–Tarski;
- the lattice viewpoint behind topologies, sigma-algebras, and domain theory.

Remark (Why this matters for the rest of the curriculum). This chapter is a breadcrumb because lattice order quietly reappears all over the place. Power sets form Boolean lattices, sigma-algebras behave like closure-stable lattice structures, open sets form a join-rich order, and fixed-point methods show up in logic, measure, computation, and analysis. Giving lattice order its own future home makes those later echoes easier to recognize.

1.2 Ordered Sets

1.2.1 Ordered Sets

Remark (Toolkit: Ordered Sets). Order is a relation, not a property. A set carries no intrinsic ordering; an order is an additional structure imposed on the set by specifying which pairs of elements stand in the ordering relation. The same underlying set can support many distinct orders simultaneously. The atomic notions developed here are:

- the **partial order** — the minimal framework in which order makes sense;
- the **strict partial order** — the asymmetric companion to the partial order, and the native language of open-ball conditions and limits;
- the **total order** — the specialisation in which every pair of elements is comparable.

Definition (Partial Order)

Definition 1.2.1 (Partial Order). A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \Longrightarrow a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \Longrightarrow a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Remark (Standard quantified statement).

$$\mathrm{PartialOrder}(S, \leq) \;:=\; \big(\forall a \in S,\ a \leq a\big) \wedge \big(\forall a, b \in S,\ a \leq b \wedge b \leq a \Rightarrow a = b\big) \wedge \big(\forall a, b, c \in S,\ a \leq b \wedge b \leq c$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{PartialOrder}(S, \leq) \;:=\; \big(\exists a \in S,\ a \not\leq a\big) \vee \big(\exists a, b \in S,\ a \leq b \wedge b \leq a \wedge a \neq b\big) \vee \big(\exists a, b, c \in S,\ a \leq b \wedge b \leq c \wedge$$

Remark (Interpretation). Order is not an intrinsic quality of a set's elements — it is an external binary relation. The three axioms together enforce that the relation be a consistent, cycle-free, directed comparison. Reflexivity says every element is at least as large as itself. Anti-symmetry is the formal engine behind squeeze arguments: if $a \leq b$ and $b \leq a$ then a and b cannot be distinct. Transitivity ensures that the ordering chains: comparisons can be composed without breaking consistency. The same set of objects can carry many distinct partial orders simultaneously — standard, reverse, or lexicographic — each defining a different relational geometry.

Remark (Dependencies).

- Partial Order

Definition

Definition 1.2.2 (Strict Partial Order)**.** A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \Longrightarrow \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \Longrightarrow a < c$.

Remark (Standard quantified statement).

$$\mathrm{StrictPartialOrder}(S, <) \;:=\; \big(\forall a \in S,\ \neg(a < a)\big) \wedge \big(\forall a, b \in S,\ a < b \Rightarrow \neg(b < a)\big) \wedge \big(\forall a, b, c \in S,\ a < b \wedge b$$

Remark (Interpretation). The strict order is the companion of the non-strict order, eliminating reflexivity and replacing anti-symmetry with the stronger asymmetry. Every strict partial order determines a partial order by $a \leq b \iff (a < b \vee a = b)$, and every partial order determines a strict partial order by $a < b \iff (a \leq b \wedge a \neq b)$. In metric spaces and analysis, strict orders are the natural language of open conditions: open balls, open intervals, and limit definitions all use $<$ rather than $\leq$ to exclude boundary contact.

Remark (Dependencies).

- Partial Order

Definition (Total Order)

Definition 1.2.3 (Total Order)**.** A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

Remark (Standard quantified statement).

$$\mathrm{TotalOrder}(S, \leq) \;:=\; \mathrm{PartialOrder}(S, \leq) \;\wedge\; \forall a, b \in S,\ (a < b) \oplus (a = b) \oplus (b < a),$$

where $\oplus$ denotes exclusive or. Equivalently, comparability suffices: $\forall a, b \in S,\ a \leq b \vee b \leq a$.

Remark (Interpretation). A partial order permits elements to be incomparable: it is possible that neither $a \leq b$ nor $b \leq a$. A total order removes this possibility entirely — every pair of elements is forced to be comparable. The rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$ under their standard orderings are totally ordered fields. By contrast, the power set $\mathcal{P}(X)$ of any set X with more than one element, ordered by inclusion, is typically not a total order: the singletons $\{a\}$ and $\{b\}$ with $a \neq b$ are incomparable.

Remark (Dependencies).

Partial order, strict partial order.

1.2.2 Chains and Antichains

Remark (Toolkit: Chains and Antichains). A poset can contain subsets whose elements are either completely aligned or entirely incomparable. These two extremes are formalised as chains and antichains. They measure the "linearity" and "width" of the order structure respectively.

Definition (Chain)

Definition 1.2.4 (Chain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \vee \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Remark (Standard quantified statement).

$$\mathrm{Chain}(C, S, \leq) \ :=\ C \subseteq S \ \wedge \ \forall x, y \in C,\ x \leq y \vee y \leq x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Chain}(C, S, \leq) \ :=\ \exists x, y \in C,\ x \not\leq y \wedge y \not\leq x.$$

Remark (Interpretation). A chain is a subset in which the partial order behaves as a total order: no two elements are left incomparable. The entire real line $\mathbb{R}$ under its standard ordering is itself a chain. Any finite totally ordered set — a ranking, a sequence of nested intervals, a linearly ordered list of stages — is a chain. Chains model the "ladder" structure of order: there is a clear direction from smaller to larger, and the path from any element to any other is unambiguous.

Remark (Dependencies).

Partial order, total order.

Definition (Antichain)

Definition 1.2.5 (Antichain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \wedge \neg(y \leq x).$$

Remark (Standard quantified statement).

$$\operatorname{Antichain}(A, S, \leq) \;:=\; A \subseteq S \;\wedge\; \forall x, y \in A,\; x \neq y \Rightarrow x \nleq y \wedge y \nleq x.$$

Remark (Negated quantified statement).

$$\neg \operatorname{Antichain}(A, S, \leq) \;:=\; \exists x, y \in A,\; x \neq y \wedge (x \leq y \vee y \leq x).$$

Remark (Interpretation). An antichain is a subset in which the partial order makes no comparisons at all: every pair of distinct elements is incomparable. In the power set $\mathcal{P}(X)$ ordered by inclusion, the collection of all subsets of a fixed cardinality k forms an antichain — no subset of size k contains another. Antichains measure the "width" of a poset: Dilworth's theorem states that the minimum number of chains needed to cover a finite poset equals the maximum size of an antichain. In higher-dimensional structures such as product orders, large antichains arise naturally and are the primary tool for understanding the lateral spread of the order.

Remark (Dependencies).

Partial order.

1.2.3 Partial and Total Maps

Remark (Toolkit: Partial and Total Maps). Before studying maps that preserve order structure, the foundational function types are recorded: a partial map need not be defined everywhere, while a total map is defined on every input. These are the base layer upon which the order-theoretic notions rest.

Definition (Partial Map)

Definition 1.2.6 (Partial Map)**.** Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\; \forall y_1, y_2 \in B, \quad (x, y_1) \in f \;\wedge\; (x, y_2) \in f \implies y_1 = y_2.$$

Remark (Standard quantified statement).

$$\operatorname{PartialMap}(f, A, B) \;:=\; f \subseteq A \times B \wedge \forall x \in A,\; \forall y_1, y_2 \in B,\; (x, y_1) \in f \wedge (x, y_2) \in f \Rightarrow y_1 = y_2.$$

Remark (Interpretation). A partial map assigns to each element of A at most one element of B. Its domain of definition $\operatorname{dom}(f) = \{x \in A : \exists y \in B,\; (x, y) \in f\}$ may be a proper subset of A. Partial maps arise in order theory when a function is well-defined only on elements satisfying some order-theoretic condition — for example, on elements that admit an upper bound.

Definition

Definition 1.2.7 (Total Map)**.** A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\; \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

Remark (Standard quantified statement).

$$\mathrm{TotalMap}(f, A, B) \;\coloneqq\; \mathrm{PartialMap}(f, A, B) \;\wedge\; \forall x \in A,\; \exists y \in B,\; (x, y) \in f.$$

Remark (Dependencies).

Partial map, Function.

1.2.4 Maps Between Ordered Sets

Remark (Toolkit: Maps Between Ordered Sets). The structure-preserving maps between posets form a hierarchy of increasing strength. An order-preserving map carries the direction of the order forward. An order-embedding carries it in both directions. An order-isomorphism is a surjective order-embedding — a perfect structural identification between two posets.

Definition (Order-Preserving Map)

Definition 1.2.8 (Order-Preserving Map)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Remark (Standard quantified statement).

$$\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \Rightarrow \varphi(x) \leq_Q \varphi(y).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \exists x, y \in P,\; x \leq_P y \wedge \varphi(x) \not\leq_Q \varphi(y).$$

Remark (Interpretation). An order-preserving map carries the direction of the order from P into Q: whenever x is at or below y in P, $\varphi(x)$ is at or below $\varphi(y)$ in Q. The implication is one-directional — order-preserving maps need not reflect the order back. Two order-preserving maps compose to an order-preserving map, making order-preserving maps the morphisms of the category of posets.

Remark (Dependencies).

Partial order.

Definition (Order-Embedding)

Definition 1.2.9 (Order-Embedding)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

Remark (Interpretation). An order-embedding both preserves and reflects the order: the order structure of P is faithfully copied into Q without distortion. Unlike mere preservation, reflection means that order-relations between images in Q genuinely originate from order-relations in P. Every order-embedding is injective (Lemma 5.2.11) and identifies P order-isomorphically with its image in Q.

Remark (Dependencies).

Order-preserving map.

Definition (Order-Isomorphism)

Definition 1.2.10 (Order-Isomorphism)**.** An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \;:=\; \mathrm{OrderEmbedding}(\varphi, P, Q) \;\wedge\; \forall q \in Q,\; \exists p \in P,\; \varphi(p) = q.$$

Remark (Interpretation). An order-isomorphism is a bijection that perfectly translates the order of P into Q and back. Two order-isomorphic posets are order-theoretically identical: any statement about P expressed purely in terms of $\leq_P$ translates to an equally valid statement about Q under φ. Order-isomorphism is the correct notion of equality for posets.

Remark (Dependencies).

Order-embedding.

Lemma

Lemma 1.2.11 (Every Order-Embedding Is Injective)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \implies \forall x, y \in P,\; \varphi(x) = \varphi(y) \Rightarrow x = y.$$

Remark (Interpretation). Injectivity follows automatically from the biconditional in the embedding definition. If $\varphi(x) = \varphi(y)$, the image is simultaneously $\leq$ and $\geq$ itself in Q; order-reflection pulls this back to P, and anti-symmetry then forces $x = y$.

Remark (Dependencies).

Order-embedding, anti-symmetry of partial orders.

1.2.5 The Covering Relation

Remark (Toolkit: The Covering Relation). The covering relation identifies the irreducible steps of a partial order: y covers x when $x < y$ and there is nothing strictly between them. In a finite poset the entire order is determined by its covering relation, which is encoded exactly by the Hasse diagram.

Definition (Covering Relation)

Definition 1.2.12 (Covering Relation)**.** Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \neg\big(\exists z \in P,\; x < z < y\big).$$

Remark (Standard quantified statement).

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \forall z \in P,\; x < z \leq y \Longrightarrow z = y.$$

Remark (Negated quantified statement).

$$\neg(x \prec y) \;\coloneqq\; x \not< y \;\vee\; \exists z \in P,\; x < z < y.$$

Remark (Interpretation). The covering relation captures the direct steps of the order — the comparisons that cannot be decomposed further. In a finite poset every strict comparison $x < y$ can be resolved into a finite chain of covers $x = x_0 \prec x_1 \prec \cdots \prec x_n = y$, so the covering relation alone determines the full order. The Hasse diagram of a finite poset is exactly the directed graph of the covering relation. In a dense order such as $\mathbb{Q}$ or $\mathbb{R}$, no element covers any other: between any two elements there is always a third.

Remark (Dependencies).

Partial order, strict partial order.

Lemma

Lemma 1.2.13 (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \iff \big(\forall x, y \in P,\; x <_P y \Leftrightarrow \varphi(x) <_Q \varphi(y)\big) \iff \big(\forall x, y \in P,\; x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y)\big)$$

Remark (Interpretation). In a finite poset the order relation, the strict order, and the covering relation each determine the others completely. This lemma shows that an order-isomorphism can be detected at any one of these three levels. The covering characterisation is the most useful in practice: since Hasse diagrams encode exactly the covering relation, two finite posets are order-isomorphic precisely when their diagrams are isomorphic as directed graphs.

Remark (Dependencies).

Order-isomorphism, covering relation.

Proposition

Proposition 1.2.14 (Hasse Diagram Characterization for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.* *Go to proof.*

Remark (Standard quantified statement).

$$P \cong Q \iff \exists \varphi : P \xrightarrow{\sim} Q,\ \forall x, y \in P,\ x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y).$$

Remark (Interpretation). The Hasse diagram is a complete invariant of the order structure of a finite poset. Checking order-isomorphism between finite posets reduces to a graph isomorphism problem on the covering graphs — two finite posets with non-isomorphic Hasse diagrams cannot be order-isomorphic.

Remark (Dependencies).

Characterizations of order-isomorphisms for finite posets, covering relation.

1.2.6 Duality

Remark (Toolkit: Duality). Every partially ordered set has a dual obtained by reversing all order relations. Any statement that holds in every poset therefore holds in every dual poset, which means its order-reversed version also holds universally. This is the Duality Principle — the formal engine behind phrases like "by duality, the corresponding result for infima follows."

Definition (Dual Ordered Set)

Definition 1.2.15 (Dual Ordered Set)**.** Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Remark (Standard quantified statement).

$$x \leq_{P^{\mathrm{op}}} y \;:=\; y \leq_P x.$$

Remark (Interpretation). Dualising a poset exchanges the roles of upper and lower: every upper bound becomes a lower bound, every supremum becomes an infimum, and every maximal element becomes a minimal element. In a finite poset the Hasse diagram of P^{op} is obtained from that of P by turning it upside down. The dual of a total order is again a total order.

Remark (Dependencies).

Partial order.

Proposition

Proposition 1.2.16 (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.* *Go to proof.*

Remark (Standard quantified statement)*.* Every universally valid statement about partially ordered sets has a universally valid order-dual statement.

Remark (Interpretation)*.* The Duality Principle eliminates the need to re-prove results whose dual is already established. Any theorem about upper bounds, suprema, maximal elements, or top elements automatically yields a theorem about lower bounds, infima, minimal elements, or bottom elements by applying the principle to P^{op}.

Remark (Dependencies)*.*

Dual ordered set.

1.2.7 Extremal Elements

Remark (Toolkit: Extremal Elements)*.* A poset may contain elements that sit at the boundary of the order. There are two distinct senses of "extreme": an element can be extreme *locally* (nothing in the poset exceeds it, though it may be incomparable with some elements) or extreme *globally* (it dominates or is dominated by every element). The definitions below make these distinctions precise, proceeding from global extremes of the whole poset, through local extremes of a subset, to the proposition that finite posets always contain the local kind.

Definition (Bottom Element)

Definition 1.2.17 (Bottom Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\bot \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \bot \leq x.$$

If a bottom element exists it is unique; it is denoted $\bot_P$ or simply $\bot$.

Remark (Standard quantified statement)*.*

$$\mathrm{Bottom}(\bot, P, \leq) \;\coloneqq\; \bot \in P \;\wedge\; \forall x \in P,\, \bot \leq x.$$

Remark (Interpretation)*.* A bottom element is below every element of the poset without exception. In $(\mathcal{P}(X), \subseteq)$ the bottom element is $\varnothing$. The natural numbers have a bottom element; the integers do not. An antichain with more than one element has no bottom element, since no member is below any other.

Remark (Dependencies)*.*

Partial order.

Definition (Top Element)

Definition 1.2.18 (Top Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Remark (Standard quantified statement).

$$\mathrm{Top}(\top, P, \leq) \;\coloneqq\; \top \in P \;\wedge\; \forall x \in P,\; x \leq \top.$$

Remark (Interpretation). Dual to the bottom element via the Duality Principle. In $(\mathcal{P}(X), \subseteq)$ the top element is X itself.

Remark (Dependencies).

Bottom element, duality principle.

Definition

Definition 1.2.19 (Maximal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Remark (Standard quantified statement).

$$\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; a \leq x \Rightarrow a = x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; \exists x \in Q,\; a < x,$$

that is, some element of Q strictly exceeds a.

Remark (Interpretation). A maximal element asserts only that nothing in Q strictly dominates it. It need not be comparable with every element of Q, so a poset can have multiple maximal elements simultaneously. In domain theory, maximal elements correspond to fully-specified or total objects.

Remark (Dependencies).

Partial order.

Definition

Definition 1.2.20 (Minimal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Remark (Standard quantified statement).

$$\mathrm{Minimal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; x \leq a \Rightarrow x = a.$$

Remark (Interpretation). Dual to maximal element via the Duality Principle.

Remark (Dependencies).

Maximal element, duality principle.

Definition

Definition 1.2.21 (Greatest Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Remark (Standard quantified statement).

$$\mathrm{Greatest}(m, Q, \leq) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; x \leq m.$$

Remark (Interpretation). A greatest element is a maximal element that is additionally comparable with every element of Q. When $\max Q$ exists, $\mathrm{Max}(Q) = \{\max Q\}$, but the converse fails: a unique maximal element need not dominate every element.

Remark (Dependencies).

Maximal element.

Definition

Definition 1.2.22 (Least Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Remark (Standard quantified statement).

$$\mathrm{Least}(m, Q, \leq) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; m \leq x.$$

Remark (Interpretation). Dual to greatest element.

Remark (Dependencies).

Greatest element, duality principle.

Proposition

Proposition 1.2.23 (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \mathrm{Max}(P)$ with $x \leq y$.* *Go to proof.*

Remark (Standard quantified statement).

$$P \text{ finite},\ Q \subseteq P,\ Q \neq \varnothing \implies \mathrm{Max}(Q) \neq \varnothing.$$

$$\forall x \in P,\ \exists y \in \mathrm{Max}(P),\ x \leq y.$$

Remark (Interpretation). Finiteness is essential. The collection of all finite subsets of $\mathbb{N}$, ordered by inclusion, is infinite and has no maximal element. In a finite poset one can always ascend from any element to a ceiling. This property is the finite shadow of Zorn's Lemma and underlies ascending chain arguments throughout algebra and order theory.

Remark (Dependencies).

Maximal element, partial order.

1.2.8 Lifting and Co-Lifting

Remark (Toolkit: Lifting and Co-Lifting). A poset that lacks a bottom or top element can be given one by adjoining a new universal bound. Lifting adjoins a bottom; co-lifting adjoins a top. These constructions are dual to each other and are used throughout domain theory, where a bottom element represents an undefined state.

Definition (Lifting)

Definition 1.2.24 (Lifting). Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp \coloneqq P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \qquad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new bottom element $\perp$ adjoined.

Remark (Standard quantified statement).

$$P_\perp = P \cup \{\perp\}, \quad \perp \notin P, \quad \forall x \in P_\perp,\ \perp \leq x, \quad \forall x, y \in P,\ x \leq_{P_\perp} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Lifting adjoins a universal lower bound to a poset that may not have one. In domain theory $\perp$ represents an undefined, divergent, or unspecified state. Any set S viewed as an antichain can be lifted to the *flat poset* $S_\perp$, in which all elements of S are incomparable above $\perp$.

Remark (Dependencies).

Partial order, Union, bottom element.

Definition

Definition 1.2.25 (Co-Lifting). Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top \coloneqq P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \qquad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new top element $\top$ adjoined.

Remark (Standard quantified statement).

$$P^{\top} = P \cup \{\top\}, \quad \top \notin P, \quad \forall x \in P^{\top},\ x \leq \top, \quad \forall x, y \in P,\ x \leq_{P^{\top}} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Dual to lifting via the Duality Principle: co-lifting adjoins a universal upper bound. The co-lifting of an antichain produces a flat poset in which all elements are incomparable below $\top$.

Remark (Dependencies).

Lifting, top element, duality principle.

1.2.9 Consistency

Remark (Toolkit: Consistency). Two elements of a poset are consistent when they share a common upper bound — when they can be jointly extended to some element above both of them. In lattice-theoretic settings this is automatic; in general posets it is a genuine condition.

Definition

Definition 1.2.26 (Consistency)**.** Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \ \wedge \ y \leq z.$$

Remark (Standard quantified statement).

$$\mathrm{Consistent}(x, y, S, \leq) \ \coloneqq\ \exists z \in S,\ x \leq z \wedge y \leq z.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Consistent}(x, y, S, \leq) \ \coloneqq\ \forall z \in S,\ x \not\leq z \vee y \not\leq z.$$

Remark (Interpretation). Consistency asserts that x and y are compatible — there is some element of S above both. In a lattice where binary joins always exist, consistency is automatic: take $z = x \vee y$. In a general poset the condition is nontrivial and may fail. In domain theory, consistency captures the idea that two partial computations describe information that can be combined without contradiction.

Remark (Dependencies).

Partial order.

1.2.10 Down-Sets and Up-Sets

Remark (Toolkit: Down-Sets and Up-Sets). A down-set is a subset of a poset that is closed downward: if an element belongs to it, everything below that element belongs to it too. An up-set is the dual notion — closed upward. Down-sets are the order ideals of lattice theory; up-sets are the order filters. The collection of all down-sets of a poset, ordered by inclusion, is itself an important poset.

Definition (Down-Set)

Definition 1.2.27 (Down-Set)**.** Let $(P, \le)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q,\ \forall y \in P, \quad y \le x \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{DownSet}(Q, P, \le) \ :=\ Q \subseteq P\ \wedge\ \forall x \in Q,\ \forall y \in P,\ y \le x \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DownSet}(Q, P, \le) \ :=\ \exists x \in Q,\ \exists y \in P,\ y \le x \wedge y \notin Q.$$

Remark (Interpretation). A down-set is a downward-closed region of the poset: once an element is included, the entire lower section below it is forced in. Down-sets generalise downward-closed intervals. The collection of all down-sets of P, denoted $\mathcal{O}(P)$ and ordered by inclusion, is itself a poset and in the finite case a distributive lattice. Birkhoff's representation theorem states that every finite distributive lattice arises this way.

Remark (Dependencies).

Partial order.

Definition (Up-Set)

Definition 1.2.28 (Up-Set)**.** Let $(P, \le)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q,\ \forall y \in P, \quad x \le y \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{UpSet}(Q, P, \le) \ :=\ Q \subseteq P\ \wedge\ \forall x \in Q,\ \forall y \in P,\ x \le y \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{UpSet}(Q, P, \le) \ :=\ \exists x \in Q,\ \exists y \in P,\ x \le y \wedge y \notin Q.$$

Remark (Interpretation). Dual to down-set via the Duality Principle: the up-sets of P are exactly the down-sets of P^{op}. In topology, filters are special up-sets closed under finite intersection and satisfying a directedness condition. The complement of a down-set is an up-set, and vice versa.

Remark (Dependencies).

Down-set, duality principle.

1.2.11 Families of Sets

Remark (Toolkit: Families of Sets). The power set of any set is naturally ordered by inclusion. Every collection of subsets — whether arising from algebra, topology, or logic — inherits this order. The key observation is that subsets correspond exactly to predicates via their characteristic functions, and that this correspondence is an order-isomorphism.

Remark (Families as ordered sets). Let X be a set. The *power set* $\mathcal{P}(X)$, consisting of all subsets of X, is ordered by inclusion: for $A, B \in \mathcal{P}(X)$,

$$A \leq B \iff A \subseteq B.$$

Any subcollection of $\mathcal{P}(X)$ inherits this order. Such a collection is called a *family of sets*.

Families of sets arise naturally when X carries additional structure. If X is a group G, the set $\mathrm{Sub}(G)$ of all subgroups and the set $\mathrm{NSub}(G)$ of all normal subgroups are each ordered by inclusion. If X is a vector space V, the set $\mathrm{Sub}(V)$ of all subspaces is ordered by inclusion. If X is a ring R, the sets of all subrings and of all ideals of R are similarly ordered. If $(X, \mathcal{T})$ is a topological space, the collections of open sets, closed sets, and clopen sets are each families of sets ordered by inclusion.

A particularly important example is the family $\mathcal{O}(P)$ of all down-sets of a partially ordered set P, ordered by inclusion.

Remark (Predicate formulation). The ordered set $(\mathcal{P}(X), \subseteq)$ admits an equivalent formulation in terms of predicates. A *predicate* on X is a function

$$p : X \to \{T, F\},$$

where T and F denote truth values. Let $\mathcal{P}^*(X)$ denote the set of all predicates on X, ordered by implication:

$$p \leq q \iff \{x \in X : p(x) = T\} \subseteq \{x \in X : q(x) = T\}.$$

Define the map

$$\Phi : \mathcal{P}^*(X) \to \mathcal{P}(X), \qquad \Phi(p) := \{x \in X : p(x) = T\}.$$

Proposition

Proposition 1.2.29 (Predicates and Power Sets Are Order-Isomorphic). *The map $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ is an order-isomorphism.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\Phi, \mathcal{P}^*(X), \mathcal{P}(X)).$$

Remark (Interpretation). Every subset of X can be represented as the truth set of a predicate, and every predicate determines a subset. The correspondence is bijective and order-preserving in both directions: $p \leq q$ if and only if the truth set of p is contained in the truth set of q, which is exactly the inclusion order on $\mathcal{P}(X)$. This identification is fundamental in logic, where sets and their characteristic predicates are treated as interchangeable, and in computer science, where sets are often implemented via their membership tests.

Remark (Dependencies).

Order-isomorphism, partial order.

Proofs

Remark (Return). Return to Theorem

Lemma (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Professional Standard Proof.
TODO: professional standard proof for order-iso-finite-characterizations. □

Detailed Learning Proof.
TODO: detailed learning proof for order-iso-finite-characterizations. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Hasse Diagram Characterization for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.*

Professional Standard Proof.
TODO: professional standard proof for hasse-diagram-characterization. □

Detailed Learning Proof.
TODO: detailed learning proof for hasse-diagram-characterization. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.*

Professional Standard Proof.
TODO: professional standard proof for duality-principle. □

Detailed Learning Proof.
TODO: detailed learning proof for duality-principle. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let* $(P, \leq)$ *be a finite partially ordered set. Then every nonempty subset* $Q \subseteq P$ *has at least one maximal element. Furthermore, for every* $x \in P$ *there exists* $y \in \mathrm{Max}(P)$ *with* $x \leq y$*.*

Professional Standard Proof.
TODO: professional standard proof for finite-poset-maximal. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-poset-maximal. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Predicates and Power Sets Are Order-Isomorphic). *The map* $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ *is an order-isomorphism.*

Professional Standard Proof.
TODO: professional standard proof for predicates-power-sets-iso. □

Detailed Learning Proof.
TODO: detailed learning proof for predicates-power-sets-iso. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Lemma (Every Order-Embedding Is Injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.*

Professional Standard Proof.
TODO: professional standard proof for order-embedding-injective. □

Detailed Learning Proof.
TODO: detailed learning proof for order-embedding-injective. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Bibliography